

ANA GAROÑA DELGADO, PhD

Molecular and Cell Biologist | Quantitative Biological Research

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PROFILE

Postdoctoral Scientist **with 9+ years of experience** cross-disciplinary research spanning genetics, genomics, cell biology, and microbiology, combining experimental research with Python/R-based computational analysis. Experienced in analysing complex genomic datasets, designing quantitative experiments and integrating computational and experimental approaches. Strong record of independent research, interdisciplinary collaboration, scientific communication and project leadership.

EDUCATION & RESEARCH EXPERIENCE

Postdoctoral Scientist

Gulbenkian Institute for Molecular Medicine (GIMM) former IGC, Lisbon, Portugal | Marco Fumasoni Lab | Jan 2023 - Jun 2026

- Led a 3.5-year project on **cell size adaptation in budding yeast** combining experimental evolution, genetic engineering and quantitative phenotyping.
- Analyzed **complex genomic datasets in Python** to identify mutations associated with cellular adaptation and stable size reduction.
- Integrated genomic, microscopy and flow-cytometry data to investigate genotype-phenotype relationships.

PhD Researcher

International Max Planck Research School for Evolutionary Biology & Kiel University, Germany | Tal Dagan Lab | Oct 2018 - Dec 2022

- Led a 4-year project on **genetic drift, plasmid inheritance and bacterial adaptation**, with implications for **antibiotic resistance spread**.
- Developed a **novel experimental system** to control plasmid allele frequency at single-cell and population levels.
- Collaborated with **mathematicians** to combine experimental data with mathematical modelling and quantitative analysis.

Earlier Research | 2013 - 2018

- M.Sc. thesis, CRG Barcelona (Ben Lehner lab), during **M.Sc. in Genetics and Genomics, Universidad de Barcelona (UB)**: Investigated gene regulatory network evolution.
- B.Sc. thesis and summer Internship, UPV/EHU (Marian M. de Pancorbo lab), during **B.Sc. in Biochemistry and Molecular Biology**: Conducted population-genetic and genomic analyses and developed a novel genetic test for lactase persistence.

SKILLS & EXPERTISE

Programming & Data Analysis: Python, R, genomic and NGS data analysis, population genetics, **Molecular Biology & Cell Biology:** Genetic engineering (including cloning and CRISPR), microscopy (brightfield, confocal, electron), flow cytometry (FACS and imaging) and experimental evolution.

Scientific Communication & Visualization: Scientific figure and infographic design (Inkscape, Adobe Illustrator), LaTeX, scientific writing and presentation.

Interdisciplinary Research: Regular collaboration with mathematicians and physicists on quantitative modelling projects and biological data analysis.

Languages: Spanish, English and Basque; Portuguese (B2); French (B1).

PUBLICATIONS

First author on 4 peer-reviewed papers in leading cell biology, genetics, microbiology and evolution journals; author on 1 additional paper. Full list available via ORCID. Selected:

- Garoña, A., Lemos, M.V., Giometto, A. & Fumasoni, M. Experimental evolution of cellular miniaturization reveals a putative mechanism for cell size evolution. *Proceedings of the National Academy of Sciences* 123 (24), e2531280123 (2026).
- Garoña, A., Hülter, N. F., Romero-Picazo, D. & Dagan, T. Segregational drift constrains the evolutionary rate of prokaryotic plasmids. *Molecular Biology and Evolution*, 38, 5610-5624 (2021).

SCIENTIFIC COMMUNICATION & LEADERSHIP

- **Scientific peer review:** Reviewed 14 research manuscripts in genetics, evolutionary and cell biology. Assessing scientific logic, methodological robustness, interpretation and clarity.
- **Training & Coordination:** Coordinated and lectured a Master's-level module in Scientific Communication and Writing, teaching students to structure, edit and present scientific work clearly and persuasively.
- **Scientific synthesis & Grant writing:** Contributed to research funding applications and a patent application, translating project objectives, preliminary data and research findings into clear narratives for non-specialist audiences.
- **Scientific visualization:** Designed infographic posters and scientific figures for projects across the research group, communicating research to both specialist and general audiences.
- **Public speaking:** Delivered research talks at 7 international conferences, including ASCB, FEMS, SMBE, and ISPB to audiences of 50-500+, and presented posters at additional scientific meetings.

REFERENCES

- **Marco Fumasoni:** group leader at the Gulbenkian Institute for Molecular Medicine (GIMM), postdoctoral advisor. marco.fumasoni@gimm.pt
- **Tal Dagan:** group leader at the Institute of General Microbiology of Kiel University, PhD advisor. tdagan@ifam.uni-kiel.de